

FiberPatrol™ FP1150

Point-locating fiber optic intrusion detection system

2021

Introducing FiberPatrol FP1150

Senstar's FiberPatrol provides protection for fence, wall, buried perimeter, as well as pipeline and data conduit Third-Party Interference (TPI) applications. It achieves this with fiber optic cables attached to a centrally located sensor unit.



FiberPatrol FP1150 Sensor Unit



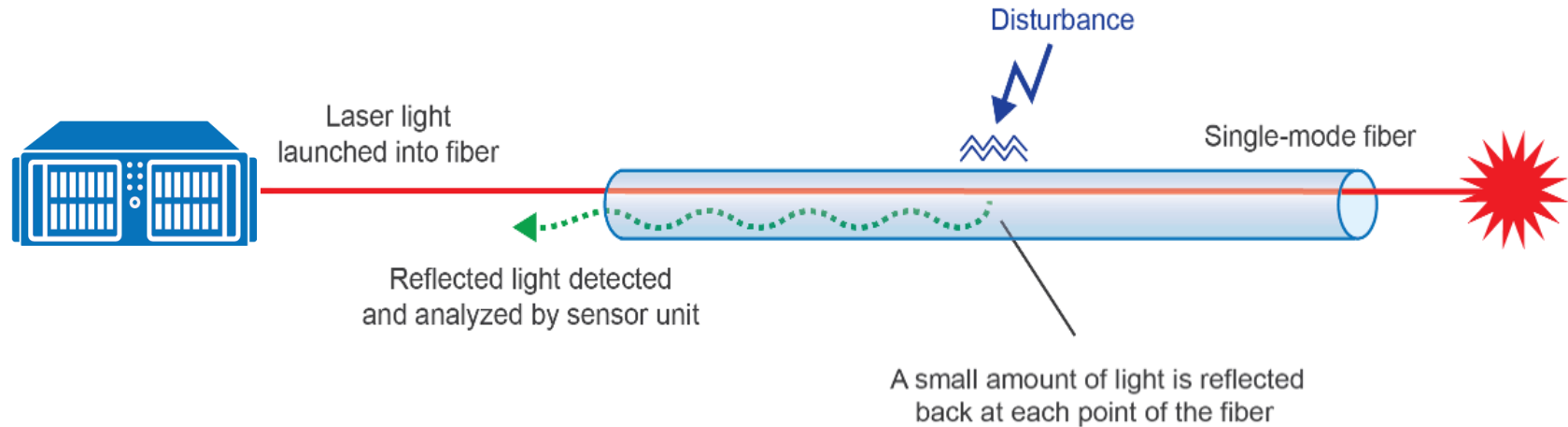
Key Benefits

- **Fiber-based** – Lightning and EMI immune, intrinsically safe in explosive atmospheres, no conductive components on the perimeter
- **Ranging** – Pin-pointing location of intrusions, software-defined zones
- **Simultaneous target detection** – Detect multiple simultaneous events
- **Environmental disaggregation** – Ranging capabilities reduce nuisance alarm by rejecting distributed weather-induced disturbances
- **Environmental compensation** – Advanced detection processing further reduces nuisance alarms
- **Cut-immune configuration** – Dual sensing channels enable a redundant-path configuration with a single sensor unit
- **Built-in SMS/VMS integration** – No separate server required
- **Redundancy** – Integration software manages redundant sensor units
- **Senstar Portfolio** – Compatible with complementary Senstar sensors for gate areas, smaller facilities, and areas requiring lighting



Principle of Operation

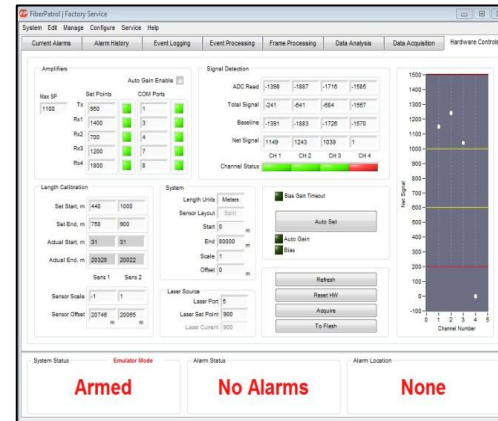
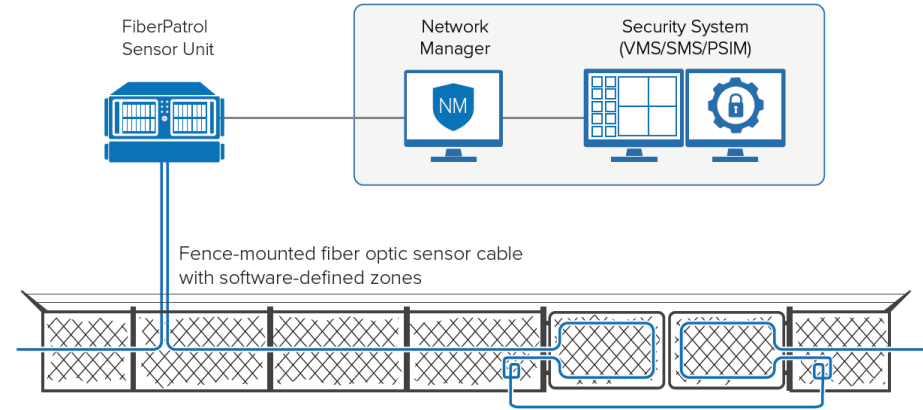
- Measures the time delay from the transmission of a pulse of laser light until the disturbance signal is reflected back to the Sensor Unit
- Disturbances can be located to within +/- 4 meters



No signal return is required from the end of the optical fiber.
If cable is cut, detection is still provided up to the point of the cut.

Basic Site Configuration Example

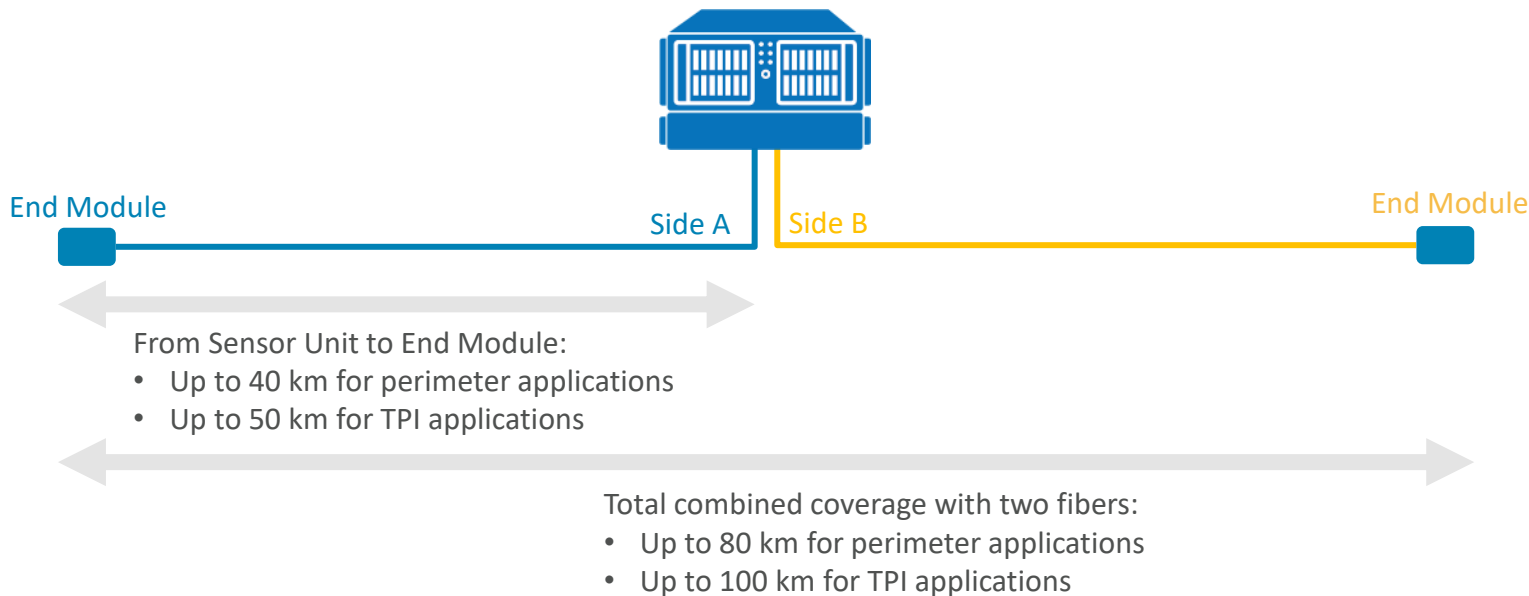
- FiberPatrol communicates with the site's security system via the Network Manager software
- Sensor Unit runs Windows OS and includes configuration software and Network Manager services
- Intrusion location of alarms and cable cuts are reported by zone, down-cable meter, and GPS (if configured)
- Sensor Unit configuration is usually performed from local rack-mounted KVM or remote desktop



FiberPatrol FP1150 Configurations

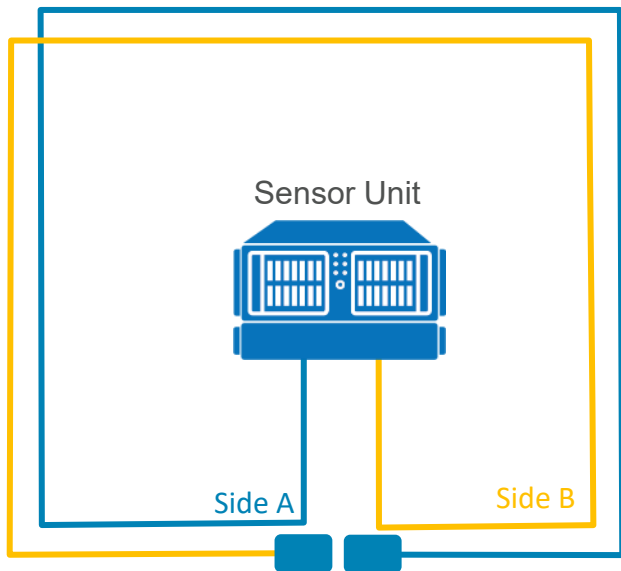
Sensor Channel Capabilities

Each FP1150 Sensor Unit supports two fibers (side A and B)



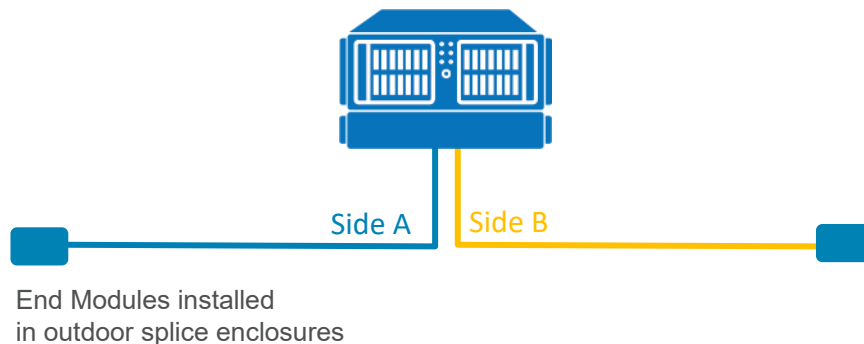
Cut-Immune and Independent Configurations

**Cut-immune configuration
("Redundant Loop")**



End Modules may be installed along perimeter in outdoor splice enclosures (most typical) or installed inside the Sensor Unit's Fiber Connection Module

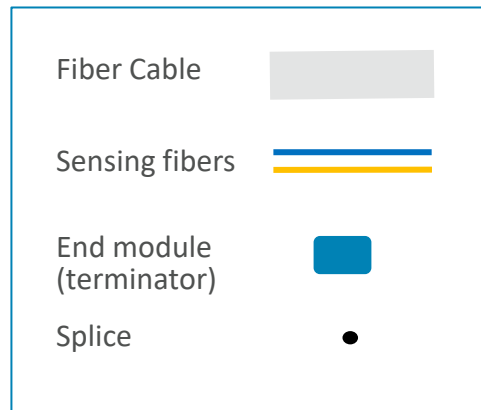
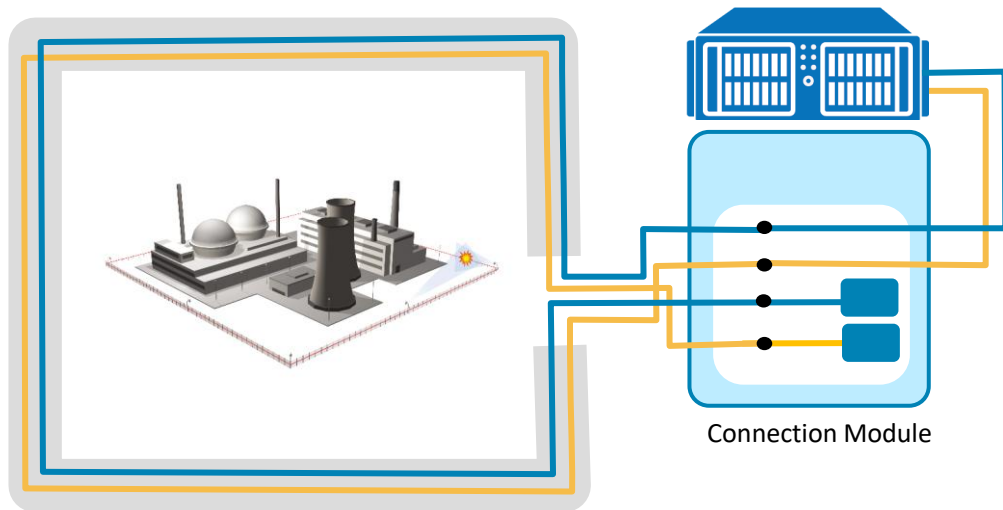
**Independent-sensor configuration
("Split")**



End Modules installed
in outdoor splice enclosures

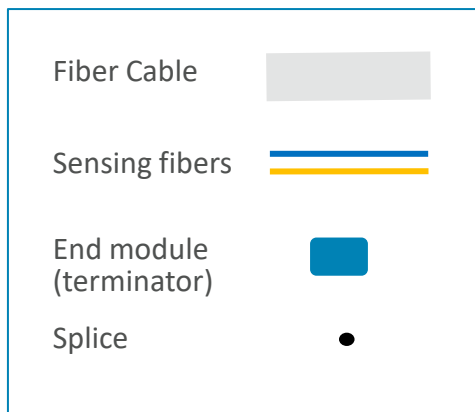
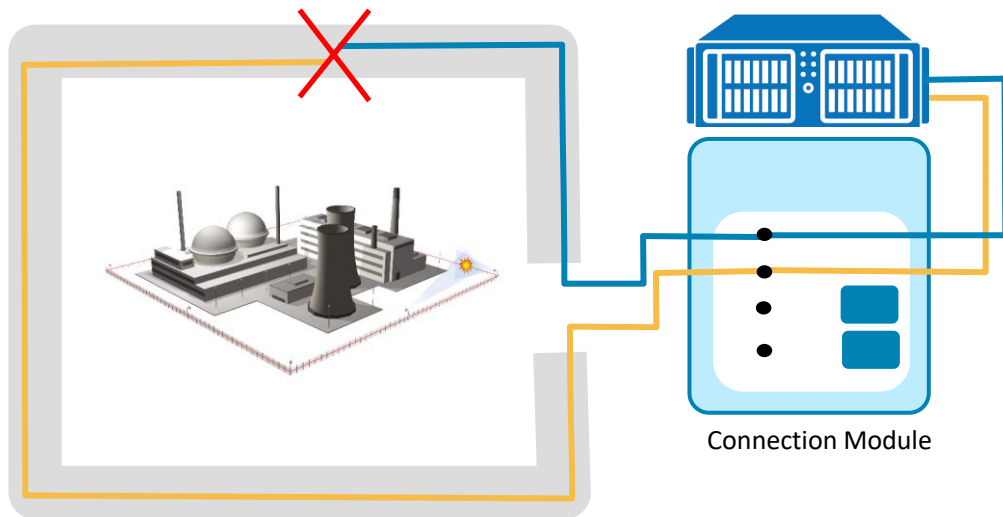
FiberPatrol FP1150 in Cut-Immune Configuration

The maximum perimeter distance that can be covered with one FiberPatrol FP1150 in the cut-immune configuration is 40 km



FiberPatrol FP1150 in Cut-Immune Configuration – Cable Cut

- After a single cable cut, the full perimeter is still protected by one fiber or the other
- The cut event and the location of the cut will be reported



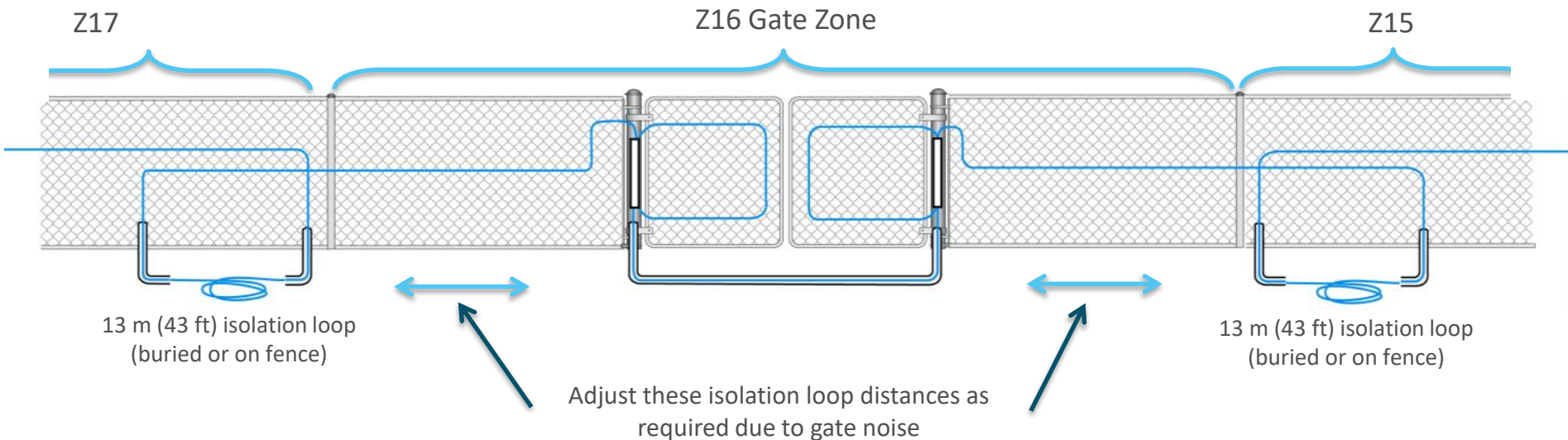
FiberPatrol FP1150 Applications

Fence Applications

- Detect climbing, cutting, or lifting of fence fabric
- Can be used on most fence types
- Up to 80 km of coverage, 40 km cut-immune
- Detection accuracy of 4 m
- Swinging gates can be protected with sensor cable
- Not recommended to install the cable on sliding gates:
 - Route sensor cable under gate area in conduit
 - Make gate area a zone and disable in software
 - Protect gate with other solution (wireless gate sensor, microwave, etc)



Swinging Gates (gate area is an independent zone)



FiberPatrol for Buried Perimeter Intrusion Detection

- Detects and locates a wide range of intrusion types:
 - Walking, running, or crawling intruders
 - Surface digging, manual and machine
 - Moving vehicles
 - Heavy vehicle engine running
 - Tunneling activities
 - Intruder or equipment movement within tunnels
- Sensor cable length up to 80 km (49.7 mi) per sensor unit
- Pinpoint intrusion locations with a ± 4 m (± 16 ft) accuracy



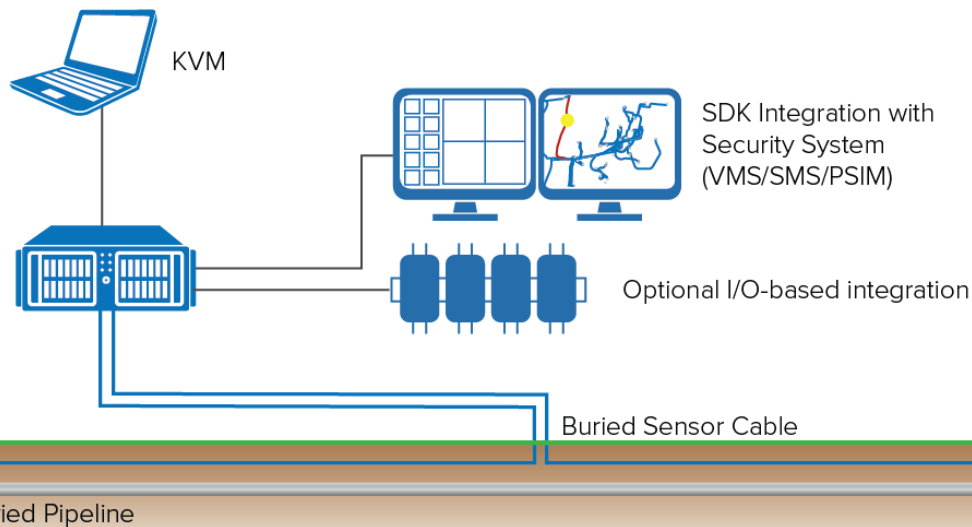
Wall-Top Applications

- Detect attempts to scale or breach masonry walls
- Detect common intruder actions:
 - leaning a ladder against wall
 - grabbing the edge of the wall to pull up
 - tossing a rope over the wall
- Sensor cable can protect the outer edge of the wall, both outer and inner edges, or the entire wall-top area



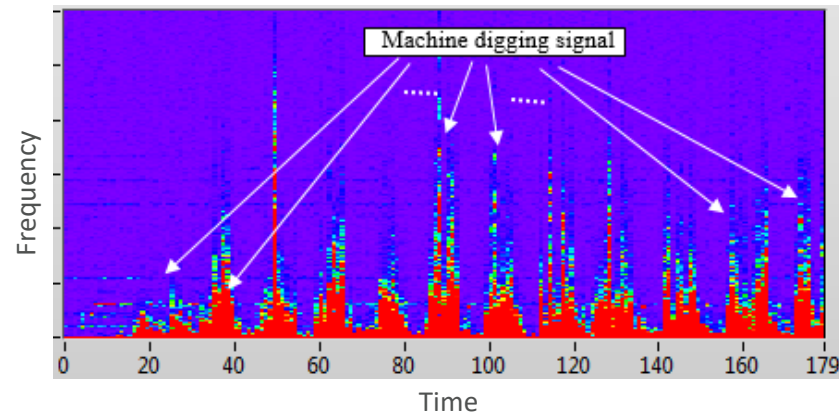
FiberPatrol FP1150 For Pipeline TPI

- Fiber optic Distributed Acoustic Sensor (DAS) system for pipeline third-party interference (TPI) detection
- Detect manual or machine digging and the presence of vehicles
- Alarms reported by zone, down-cable distance, or GPS
- Up to 100 km (62.1 mi) of coverage per processor
- No conductive components in the field, EMI and lightning immune, intrinsically safe



Typical TPI detection performance

Event	Typical detection range (perpendicular to sensor)
Manual digging	5 to 20 m (16 to 65 ft)
Light vehicle	3 to 10 m (10 to 33 ft)
Heavy excavating machines	10 to 60 m (33 to 200 ft)



Data Conduit TPI Detection Applications

- Protect data conduits against third-party interference (TPI)
- Use unused fibers from existing telecommunications-grade cable (if performance specifications are met)



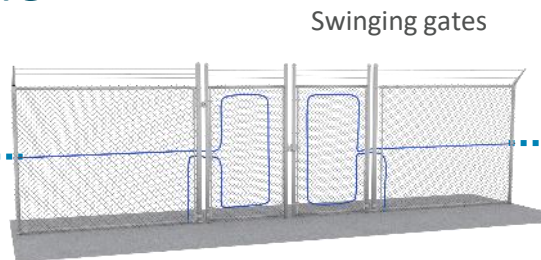
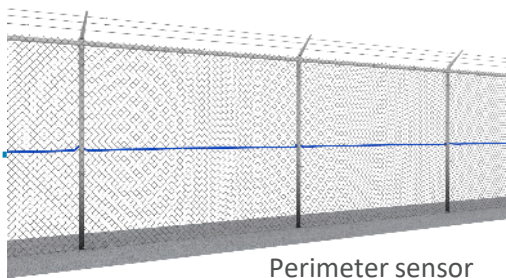
General Monitoring Applications

DAS technology can be used in other applications to enhance security:

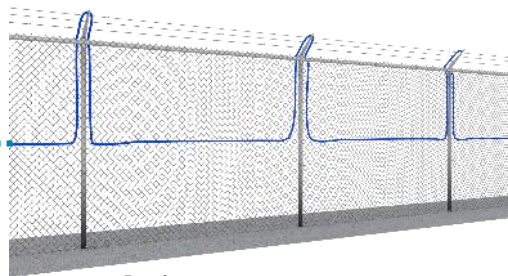
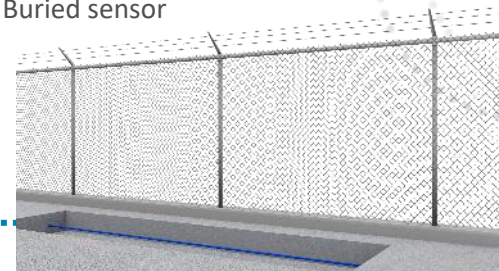
- Monitor location and speed of vehicles
- Monitoring of dividing fence to detect ingress of vehicles onto active track
- Explosion detection in vehicle tunnels



Mixed Perimeter Example



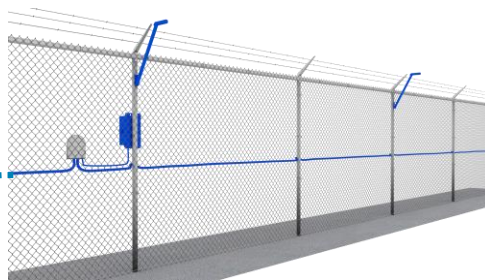
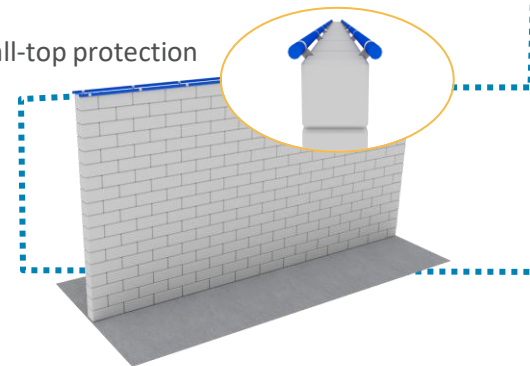
Buried sensor



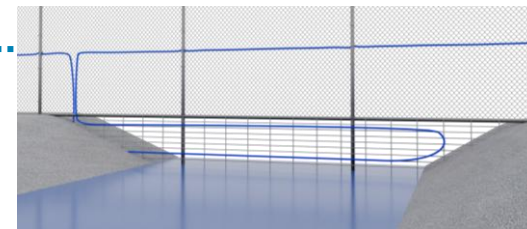
Perimeter sensor
with enhanced anti-climb detection



Wall-top protection



Section of perimeter illuminated with Senstar LM100
(use extra sensor cable fibers for comms)



Culvert protection

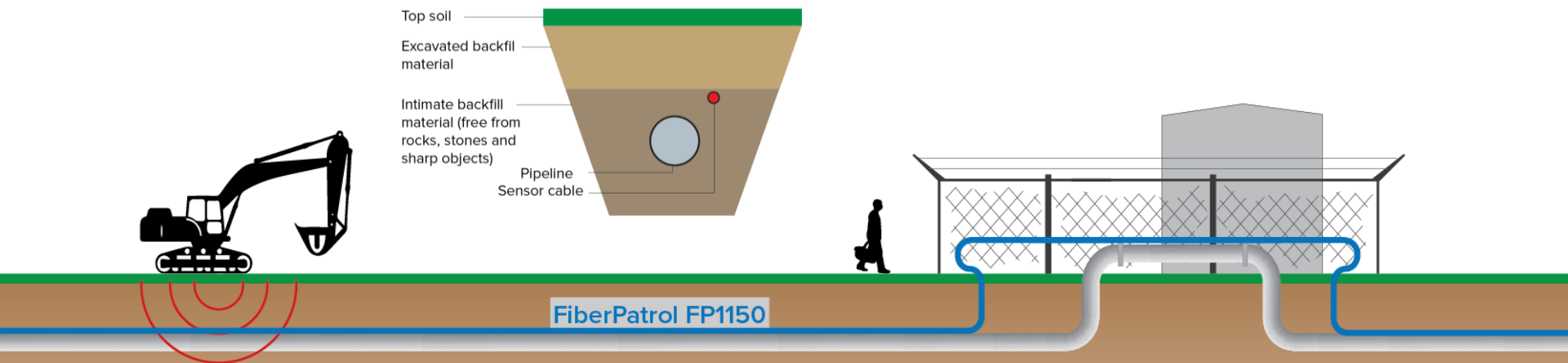
Pipeline Application Example – TPI and Fence Protection Combined

Pipeline Third-Party Interference

- Manual or machine digging near pipeline
- Light vehicles near pipeline
- Heavy excavating machines
- Hot tapping and other direct interference

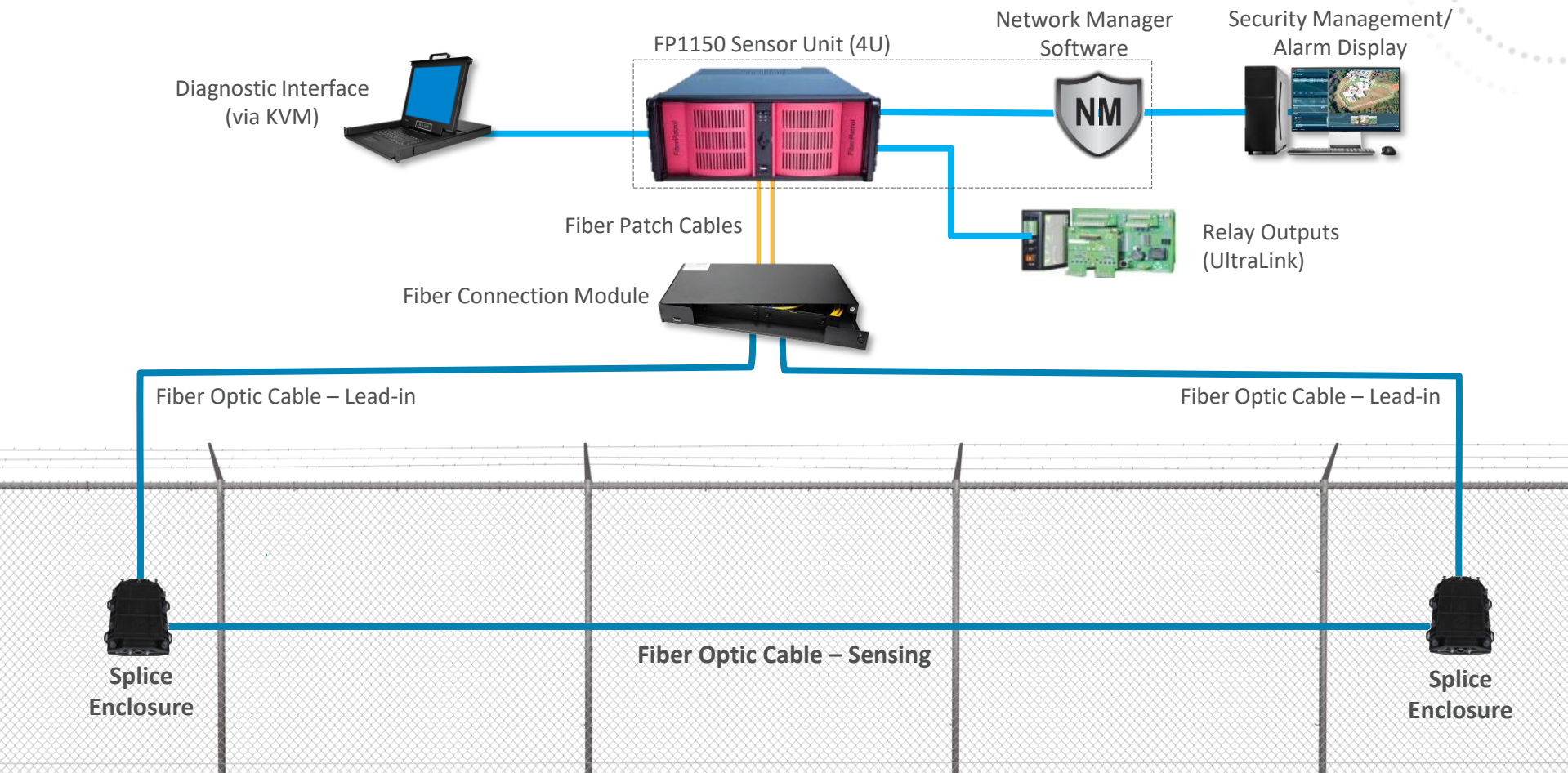
Intrusion Detection for Above-Ground Assets

- Attempts to cut, climb, or break fence fabric
- Activity near perimeter



FiberPatrol Major System Elements

Components Pictorial



FP1150 Sensor Unit

- Performs all optical and digital processing
- Two FC/APC optical connectors for field fiber attachment
- Dual-redundant hot-swap solid-state hard drives, RAID1
- Dual-redundant hot-swap PSUs
- Dual Gigabit Ethernet ports
- AC Power: 100–240V, 50/60Hz
- 4U rack-mount unit
- Windows 10 Pro 64-bit



Fiber Optic Cable

- Single mode fibers
- Loose tube, gel-filled, water blocking tape
- 0.24 db/km attenuation @ 1550 nm (installed) for 80 km coverage distance
- 0.28 db/km attenuation @ 1550 nm (installed) for 72 km coverage distance
- Central strength member, rip cord
- Fiber count varies from 12 to 60 fibers
- Outside diameter: 11.1 mm (0.44") typical
- Bend radius
 - Dynamic: 20X outside diameter (22 cm)
 - Static: 10X outside diameter (11 cm)
- Tensile rating
 - Installation: 2,700 N (600 lbf)
 - Residual: 800 N (180 lbf)
- Crush resistance
 - Short-term: 220 N/cm (125 lbf/in)
 - Long-term: 110 N/cm (63 lbf/in)
- Temperature
 - Operation: -40 to 70°C (-40 to 158°F)
 - Installation: -30 to 60°C (-22 to 140°F)

MDPE Outer Jacket

Water Blocking Tape

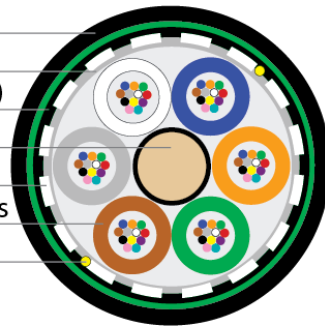
MDPE Inner Jacket (Double Jacket Designs Only)

Central Strength Member

Outer Strength Members (where applicable)

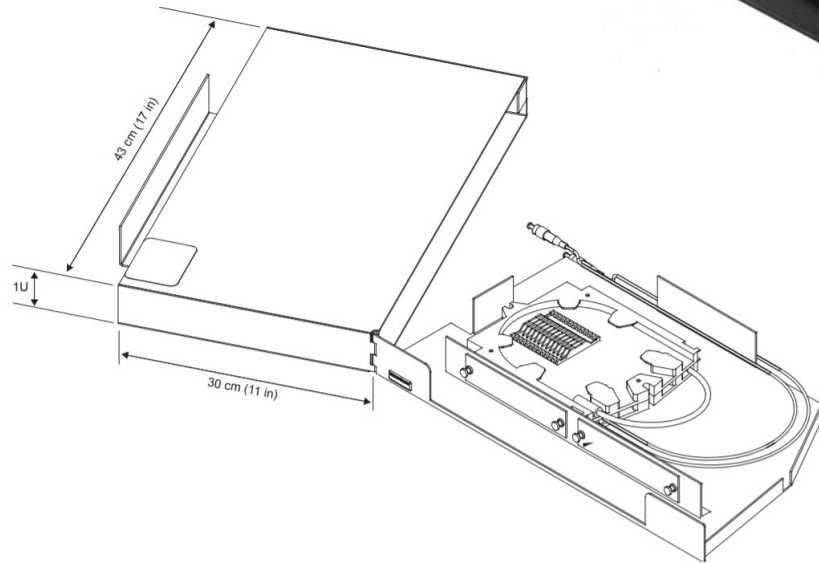
Gel-Filled Buffer Tube Containing Up To 12 Fibers

Ripcord



Fiber Connection Module

- 1U 19-inch rack-mount enclosure
- Houses the dual end modules
- Enables the splicing of field cable to patch cords (for connection to the FP1150 sensor unit)



Outdoor Splice Enclosure

- Part Number: GM0749-24
 - 3 port
 - Environmentally sealed
 - 2 splice-tray capacity

- Part Number: GM0749-48
 - 3 port
 - Environmentally sealed
 - 4 splice-tray capacity



Exterior view

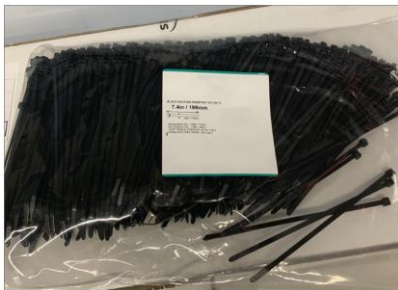


Interior view

Splice Enclosure and Consumables



Cable Ties and Clamps



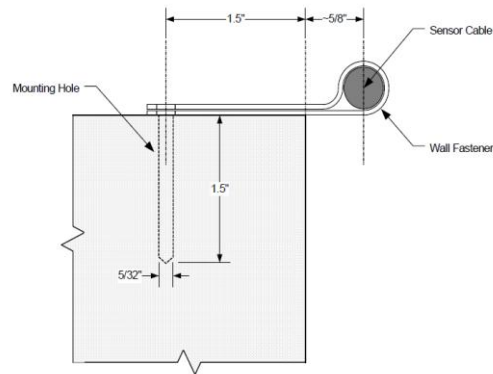
Nylon-12 cable ties

- Attach cable to fence fabric
- Sold in packages of 1000
- 12–15 year life expectancy



Stainless Steel cable ties

- Tamper-resistant
- Attach cable to fence fabric
- Sold in packages of 100
- Use tool to simply attaching/cutting



Wall clamp

- Attach cable to walls

End Module

- Wherever a fiber ends, an End Module is installed
- Eliminates reflections and allows detection right up to the end of the fiber
- In the cut-immune configuration End Modules are normally installed inside the Fiber Connection Module
- In the split configuration, each End Module is normally installed inside a splice enclosure



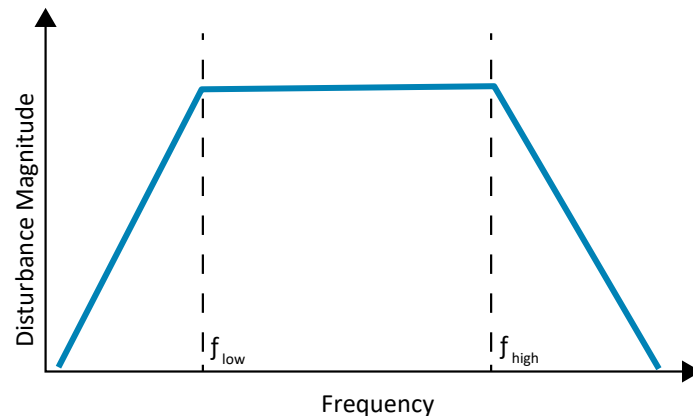
FiberPatrol Detection Processing

Detection Overview

- FiberPatrol employs multiple processing stages to discriminate between incidental disturbances and events of interests (intrusions, etc.)
- Detection settings include:
 - Frequency response range
 - Disturbance threshold
 - Disturbance life ("accumulation window")
 - Alarm threshold (frame count threshold)
 - Cut count for fence-cut detection
 - Disturbance Duration threshold

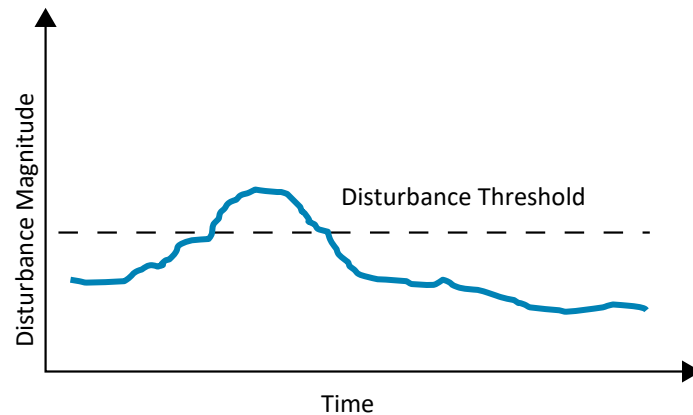
Frequency

- FiberPatrol is highly sensitive to vibrations of the sensing fiber which results in slight changes in the intensity of the light returned from the point of the disturbance
- Frequency range of the variations in the light intensity varies depending on the application and medium (fence vs. buried, type of fence, etc.)
- FiberPatrol allows setting the lower and upper cut-off frequencies to optimize system performance
- Frequency filtering enables the system to compensate for differences in fence construction and weather conditions



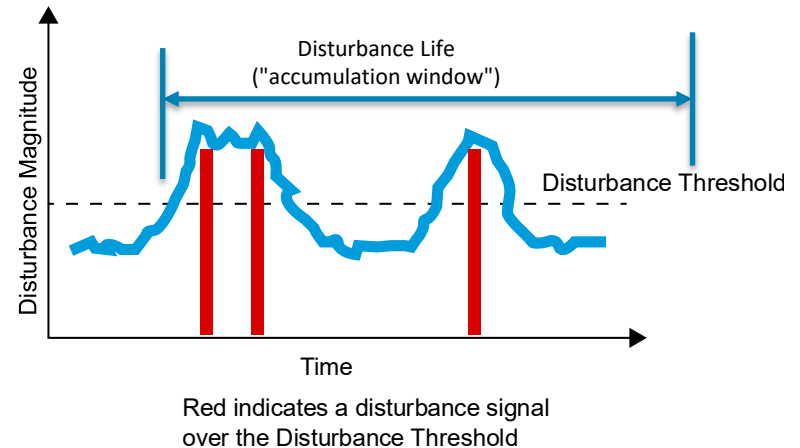
Disturbance Threshold

Only disturbance signals that exceed the Disturbance Threshold are processed further



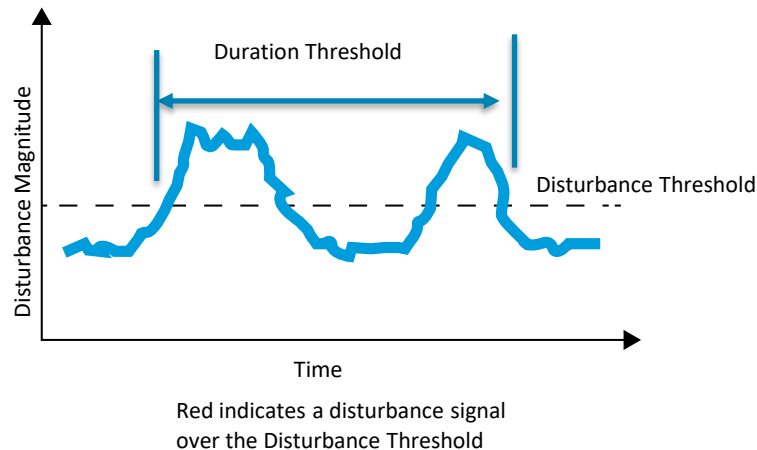
Disturbance Counts and Alarm Threshold

- Disturbances that exceed the Disturbance Threshold during a system frame add to the Disturbance Count
- The number of counts is a measure of the overall energy of the disturbance
- The configured number of disturbances required to register as an alarm is called the Alarm Threshold
- The Disturbance Count must reach the Alarm Threshold within a set amount of time called the “Disturbance Life”



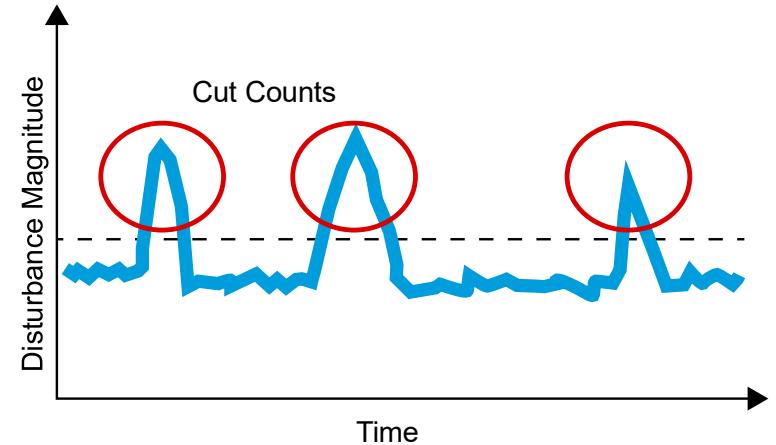
Duration Threshold

- Minimum event duration (in seconds) required in order to declare an alarm
- Used to reject events that are too brief to be considered a valid intrusion attempt (slamming of gate, soccer ball against fence, etc.)



Fence Cut Counts

- Counting cuts is a specialized way to process the disturbance signal that is optimized to detect stealthy cutting of the fence fabric
- With a sharp set of cutters the width of any one cut can be very narrow
- A knowledgeable intruder may space out their cuts to try to have the system “time out” between cuts



Detection Settings for Fence Climb and Fence Cut

FP1150 provides separate settings for fence climb (intrusion) and fence cut

☒ Intrusion ☐ Fence Cut

Frame Duration 2: 200 ms

Filter Cutoffs

Low Frequency 150 Hz

High Frequency 1000 Hz

Intrusion Detection Thresholds

Disturbance Threshold 8

Alarm Threshold 3

Duration Threshold 2 s

Intrusion Detection Window

Disturbance Life 10 s

Disturbance Mask 0.3 s

Event Detection

Disturbance Range 24 m

Event Life 15 s

☐ Intrusion ☒ Fence Cut

Frame Duration 2: 200 ms

Filter Cutoffs

Low Frequency 150 Hz

High Frequency 1000 Hz

Fence Cut Detection Thresholds

Cut Distb Threshold 11

Cut Alarm Threshold 4

Fence Cut Detection Window

Cut Window 20 s

Cut Profile 0.3

Cut Distb Mask 1 s

Event Detection

Disturbance Range 24 m

Event Life 15 s

Common settings

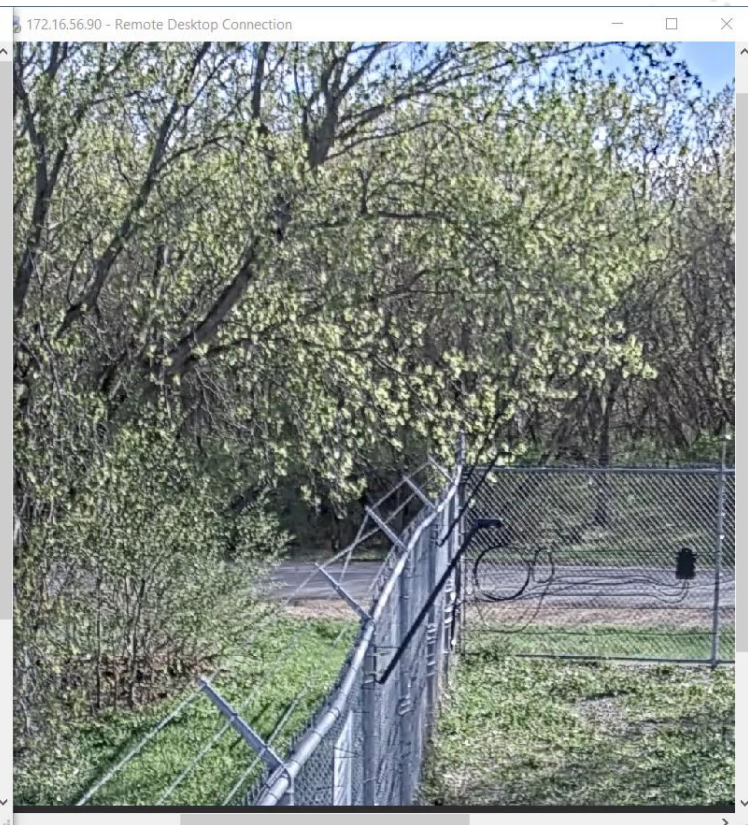
Separate setting for fence climb (intrusion) and fence cut

Common settings

Fence Climb Showing System Response

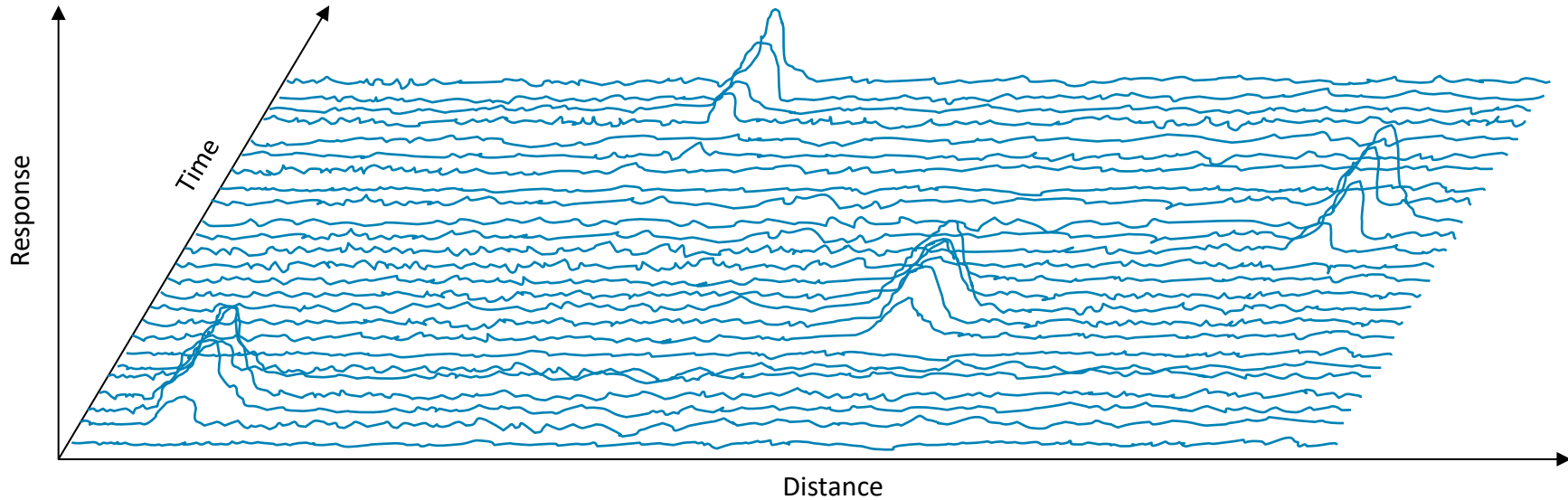


Fence Cut Showing System Response



Disturbance Example - Wind

Wind Gusts



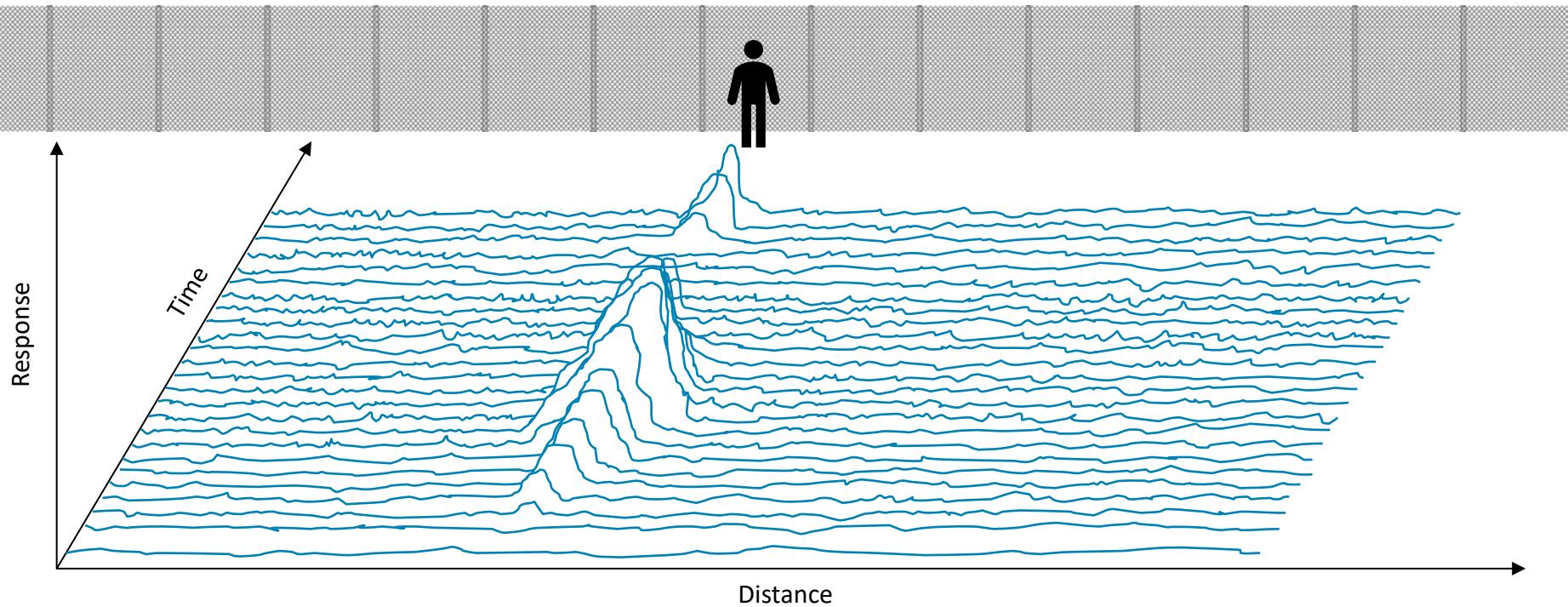
Wind tends to create a series of disturbances randomly distributed along the length of the fence – none meet the threshold conditions.

Play

Reset

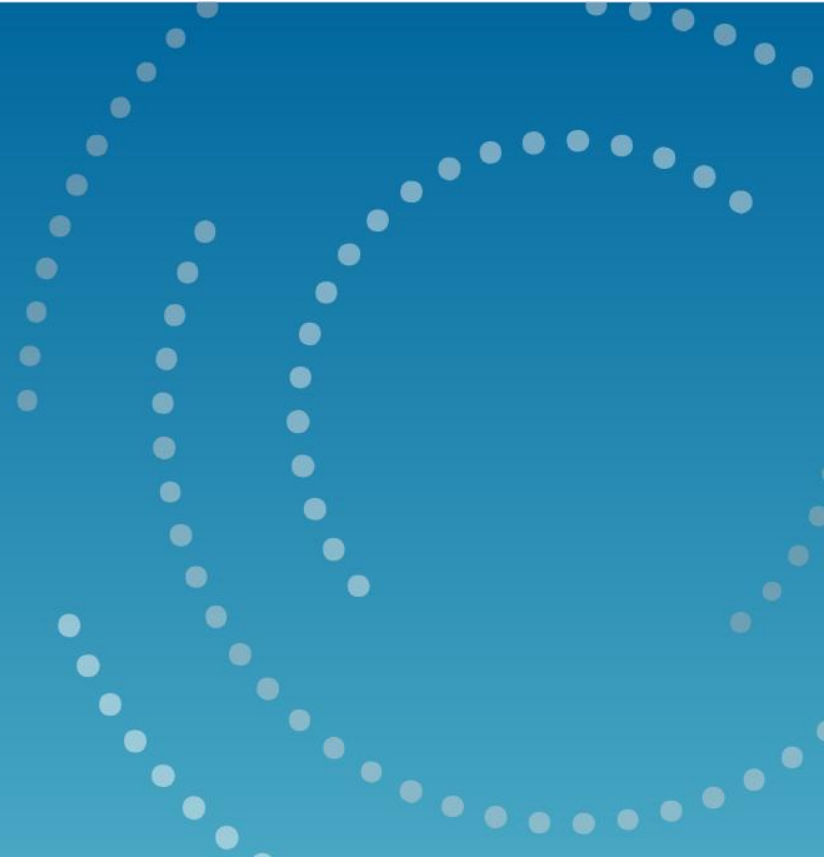
SENSTAR

Disturbance Example – Real Intrusion

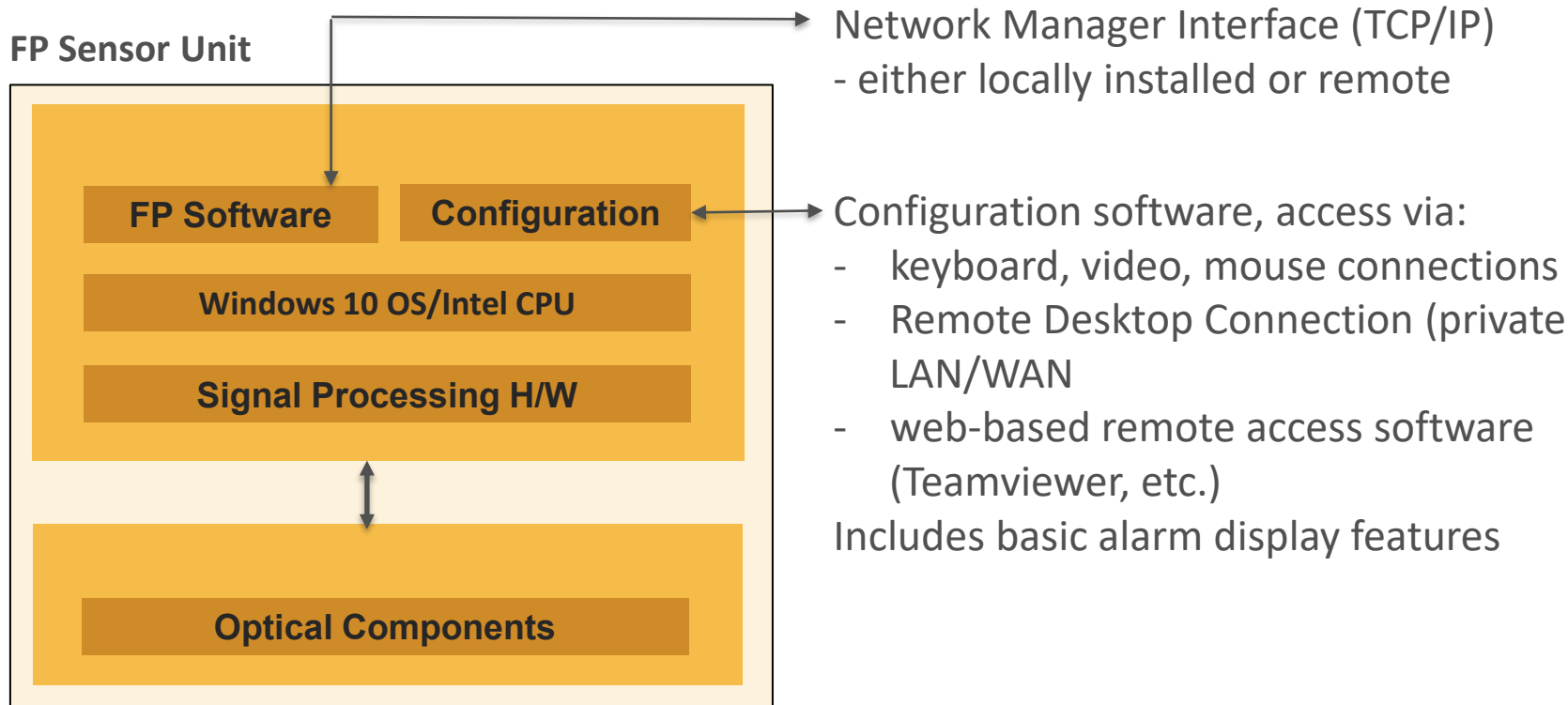


An actual intrusion creates a succession of disturbance signals concentrated at one location – meets threshold conditions.

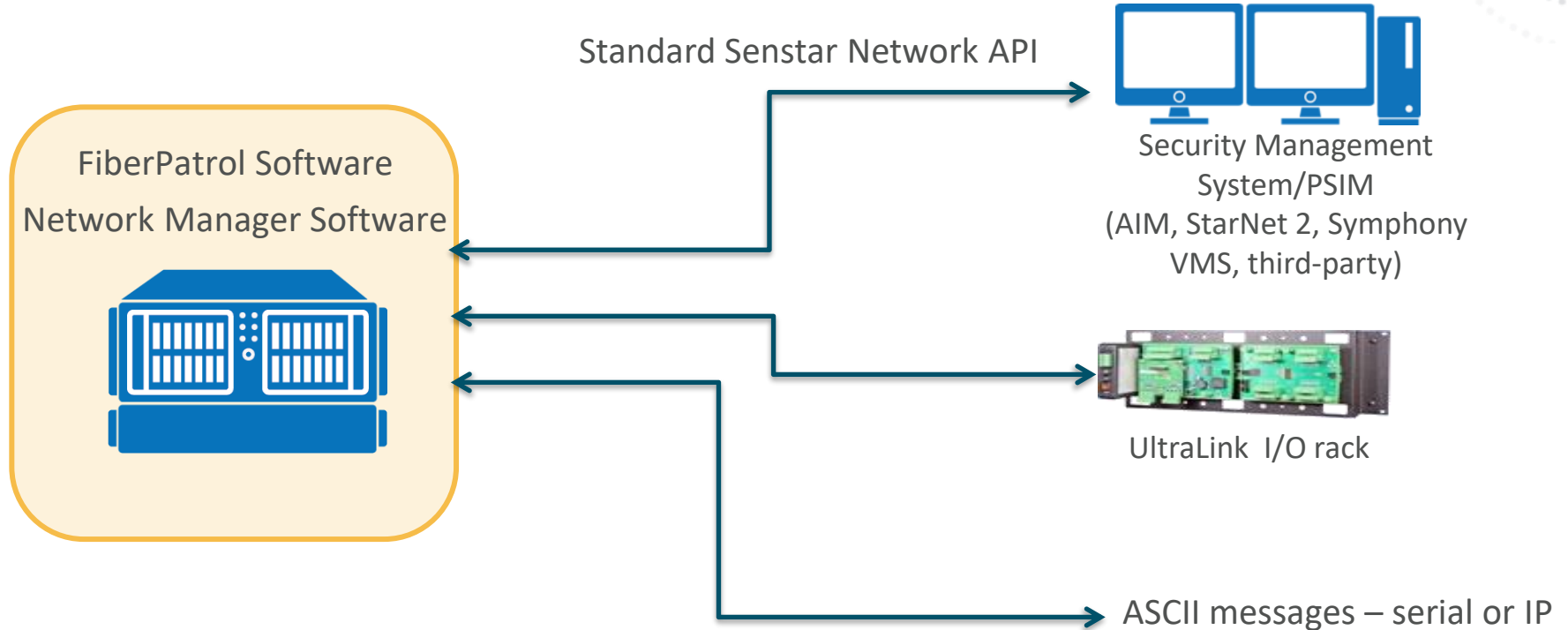
FiberPatrol Integration



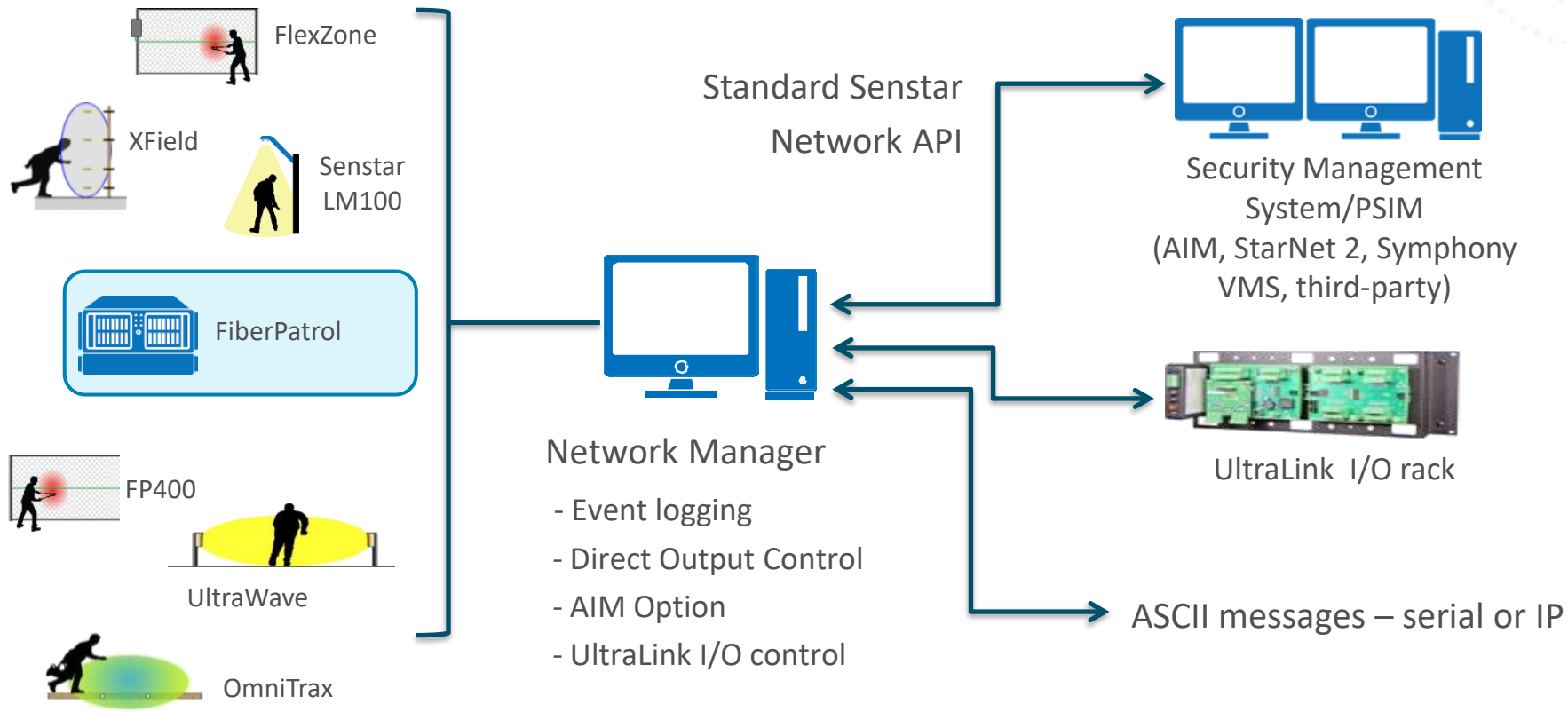
Software Interfaces



Network Integration



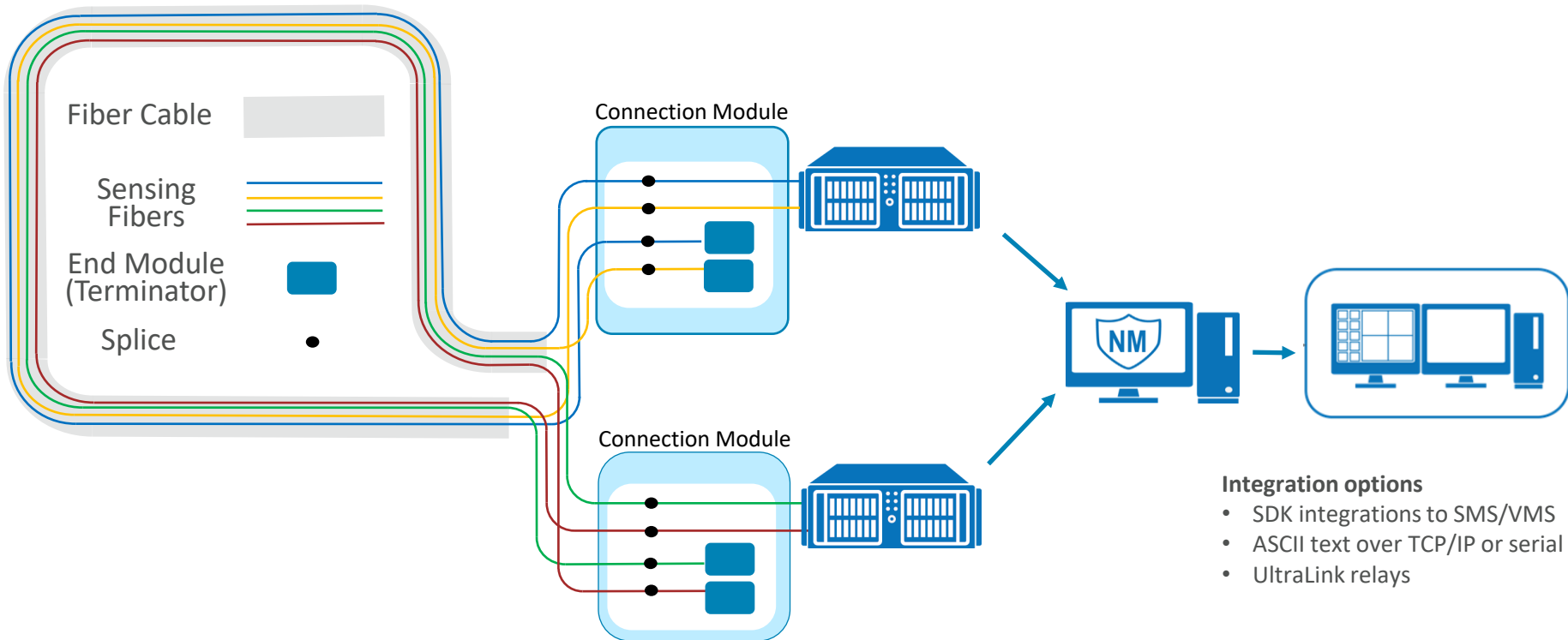
Interface Compatibility



FiberPatrol FP1150 Sensor Unit Redundancy

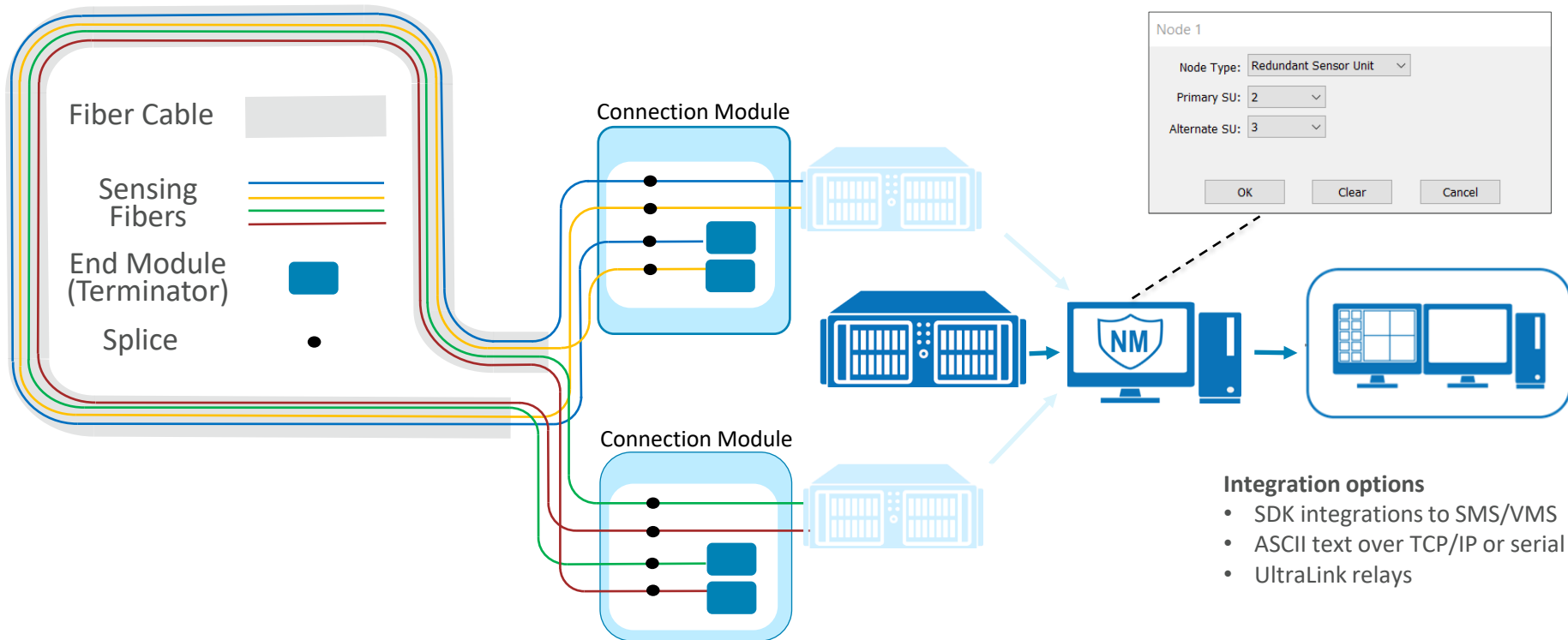
Redundant Sensor Units

For critical applications redundant sensor units can be used in parallel to maintain system operation even after if Sensor Unit should fail.



Redundant Sensor Units

The Senstar “Redundant Sensor Unit” software construct makes the two FP1150 units appear as one to the Security Management System (SMS)



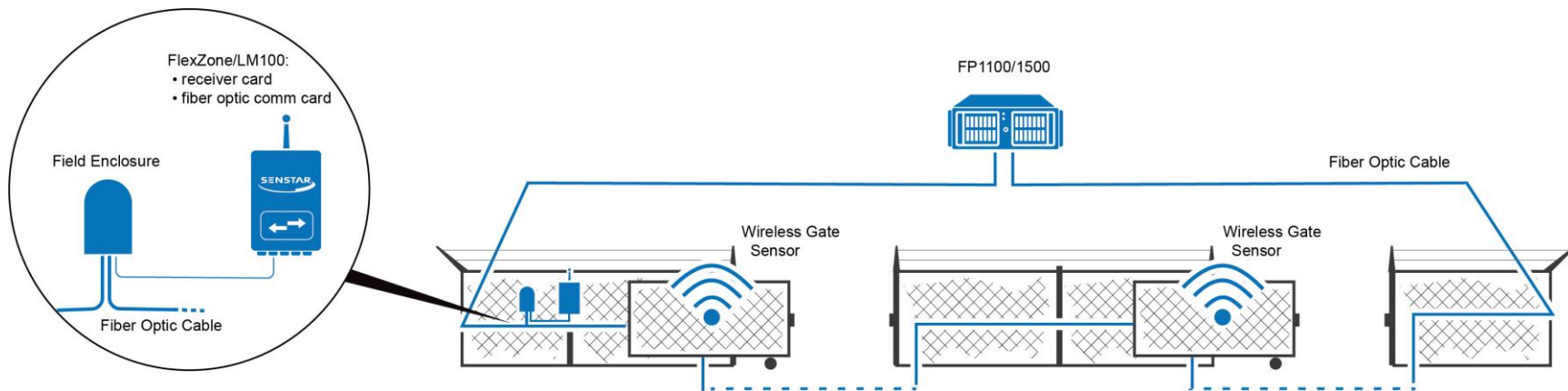
FiberPatrol FP1150 – Complementary Senstar PIDS Products

FiberPatrol FP1150 – Complementary Senstar PIDS Products

- FP400
- LM100
- FlexZone
- Wireless Gate Sensor
- UltraWave

Use of Wireless Gate Sensor with FiberPatrol

- Either LM100 or FlexZone can be used as a base station for the Wireless Gate Sensor
- Fibers in the FiberPatrol cable can be used as the Silver Network communications path for the LM100 gateway or FlexZone processor



FiberPatrol Summary

Summary

- Long-range fiber optic sensor for fence, wall, dataline and pipeline applications
- Accurately identifies the location of attempted intrusions
- Detects and accurately locates multiple simultaneous threats
- Cut-immune configuration provides cut immunity
- Centralized processor – no electronics, power or grounding required on the perimeter
- Immune to lightning and EMI
- Can be integrated with other security and assessment systems allowing efficient centralized command and control



Product Documentation



Product Video



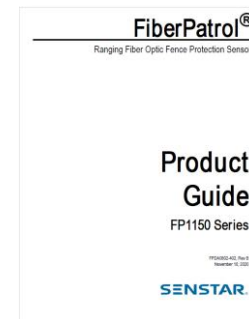
Datasheets (1 per application)



Component Technical Specifications
(14, + more to come)



A&E Specification



Product Guide

Thank You

Backup Slides

FiberPatrol Price List Elements

Progressive Pricing Model

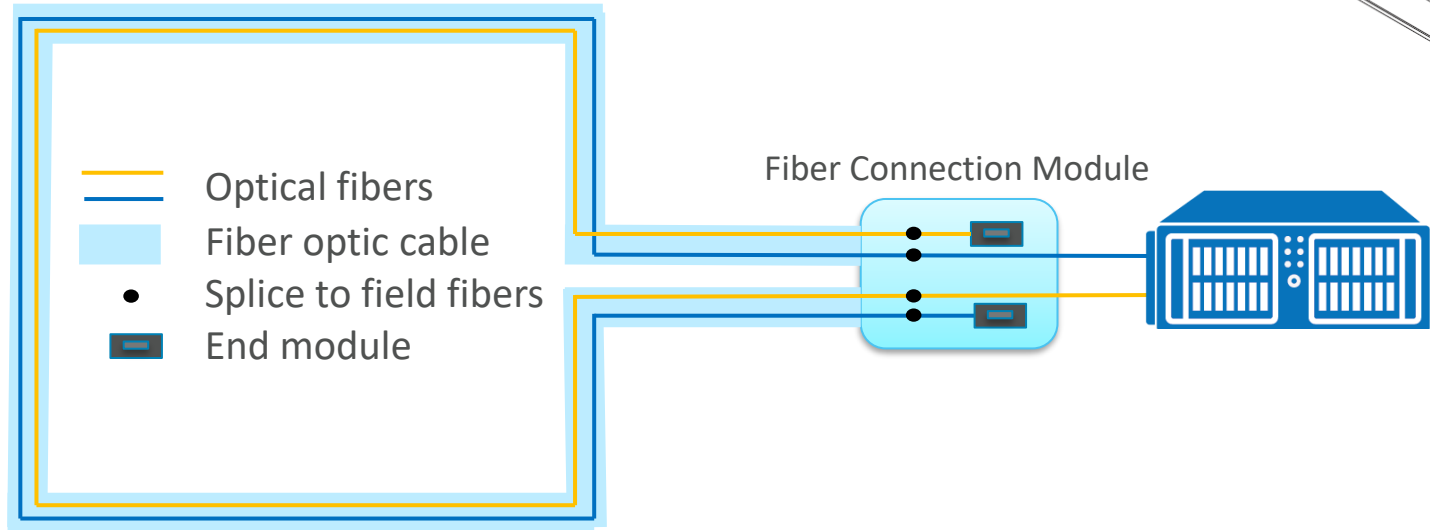
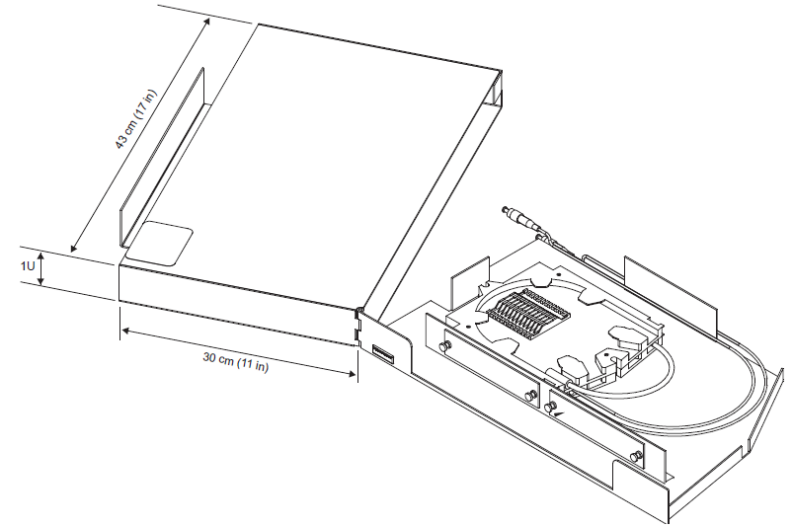
Senstar offers an innovative and unique per-meter pricing model:

- minimize pricing discontinuities across distances
- provide certitude to customers on the possibility of extending the length capability of their system
- simplifies stocking of spares

FP115005U	FP1150 unlicensed Sensor Unit for fence, wall, buried perimeter, and TPI applications. Hardware capable of providing up to 5 km (3.10 mi) of detection processing on each of its two sensor channels, up to 10 km (6.21 mi) in total. Requires separately-purchased activation license FP-PML-05 for detection processing.
FP-PML-05	Per-meter activation license applicable to FP115005U Sensor Unit.
FP115040U	FP1150 unlicensed Sensor Unit for fence, wall, buried perimeter, and TPI applications. Hardware capable of providing up to 40 km of detection processing on each of its two sensor channels, up to 80 km in total. 100 km total for TPI applications. Requires separately-purchased activation license FP-PML-40 for detection processing.
FP-PML-40	Per-meter activation license applicable to FP115040U Sensor Unit.

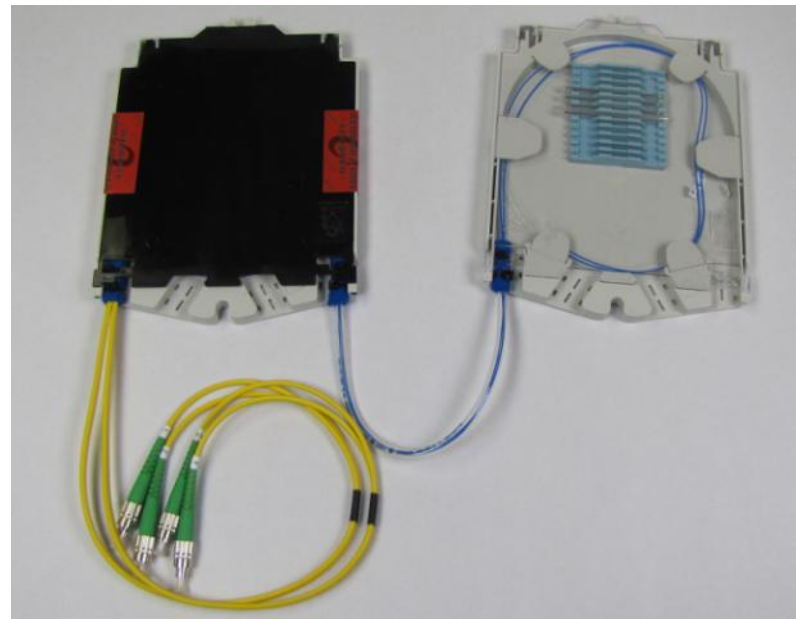
Fiber Connection Module

- Part Number: FPMA0922
- Used with FP1150



Dual End Module

- FPMA0222
- Optical modules in one sealed splice tray, field connections made to second tray



Buried Vault

- Part Number: GM0748
 - 100 x 75 x 45 cm
 - 30 x25 x 18 in



Replacement Disk Drives

- FPPG0201-001
 - Replacement blank 250 GB solid-state disk drive for FP1150 in hot-swap tray
- FPPG0202-001
 - Pair of replacement blank 250 GB solid-state disk drives. Software installation and setup sold separately



Replacement PSU

- GB0390
- replacement power supply module for FP1150 Sensor Unit



Components

Sensor Unit Accessories	
GB0296-17	17 in 1U Rackmount KVM (KB/LCD/Mouse)
FPKT0400	8 Port KVM switch w/ 2 sets of cables
FPMA0922	FiberPatrol fiber connection module for FP1150 systems. Includes two patch cords, two end modules, associated splice trays, and 1U rack-mount splice enclosure
FPMA0222	Dual End module for FP1150 systems

Components

Field Accessories	
GM0749-24	Field Splice Enclosure (24 splice capacity, 3 cable ports)
GM0479-48	Field Splice Enclosure (48 splice capacity, 3 cable ports)
FPKT0200	Splice Consumables kit
GH1080-08	3/16" x 08" (4.8 x 20.3 cm) stainless steel cable ties (100ea)
GH1080-08C	3/16" x 08" (4.8 x 20.3 cm) black-coated stainless steel cable ties (100ea)
GX0310	Tool - manual tension and cut-off tool for stainless steel cable ties
FPKT0300	FiberPatrol wall fastener (100ea)
FPKT0500	Sensor cable management kit for gates. One meter (3 ft.) of 5 cm. (2 in.) diameter split conduit, 2.3 meters (7.5 ft.) of split cable loom, two (2) hose clamps
GX0314	Fiber optic technician's toolkit
GX0321	Fiber optic connector cleaning Kit

Components

FiberPatrol Cable	
FPSP0412	Fiber optic sensor/lead cable for fence applications, 12 fibers
FPSP0424	Fiber optic sensor/lead cable for fence applications, 24 fibers
FPSP0436	Fiber optic sensor/lead cable for fence applications, 36 fibers
FPSP0448	Fiber optic sensor/lead cable for fence applications, 48 fibers
FPSP0624	Fiber optic sensor/lead cable for fence and buried applications, 24 fibers, corrugated steel tape armor, double polyethylene jacket
FPSP0648	Fiber optic sensor/lead cable for fence and buried applications, 48 fibers, corrugated steel tape armor, double polyethylene jacket

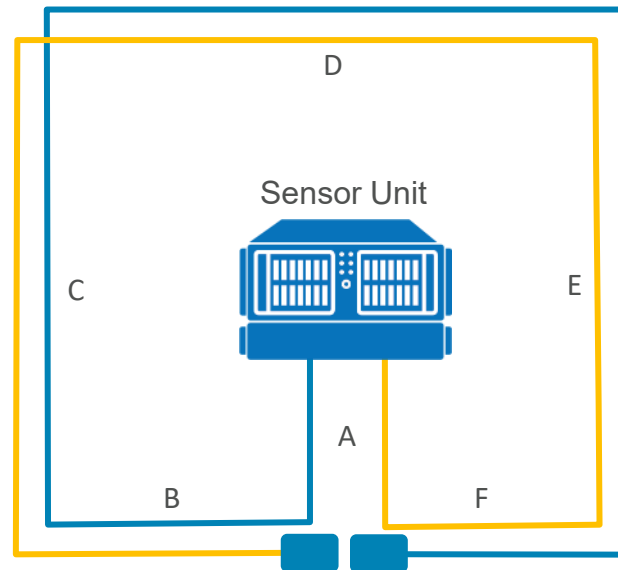
Integration Software	
00FG0220-XXY	Network Manager software
00SW0240-XXY	Network Manager – LenelS2 OnGuard Gateway
00SW0261-XXY	Network Manager – Genetec Security Center Gateway
00SW0280-XXY	Network Manager – Milestone XProtect Gateway

Configurations and Optical Length

FP1150 Redundant Loop Optical Lengths

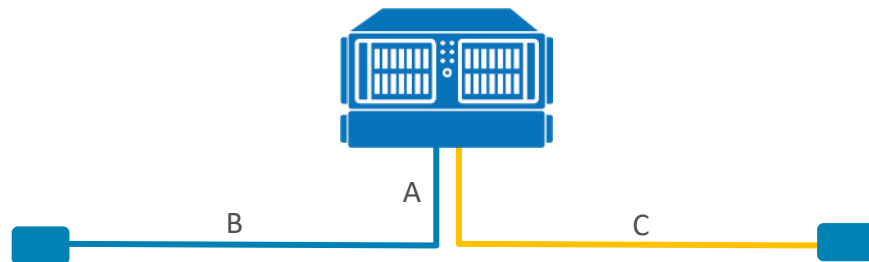
This is the most typical configuration for securing a closed perimeter:

- Budget 20% extra over perimeter length
- Install end modules on perimeter to avoid licensing the lead-in/leadout section
- $A + B + C + D + E + F$



FP1150 Independent-Sensor Configuration

- Optical path = longer of (A + B) or (A + C)
- Budget 20% extra for perimeter length or 5% for TPI detection
- Optical path of each sensor channel must be less than 40 km (perimeter) or 50 km (TPI detection)



Key Specifications

	Perimeter	Pipeline or Data Conduit TPI
Coverage distance - Independent sensor configuration	Up to 80 km (49.7 mi)	Up to 100 km (62.1 mi)
Coverage distance - Cut-immune configuration	Up to 40 km (24.85 mi)	Up to 50 km (31.06 mi) (using overlapping sensor cables from adjacent Sensor Units)
Location accuracy	4 m (13 ft)	±4m (±13 ft)
Virtual zones	Up to 1,440	Up to 1,440
Probability of detection (Pd)	95%	95%
False alarm rate	1 / km / year	1 / km / year
Nuisance alarm rate	Low	Low



Thank You.